



Lecture 3: Stability

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Poles and Zeros

- ❑ Pole, zero, and gain defined by Douglas:

$$G(s) = \frac{s + 5}{(s + 1)(s + 3)}$$

Poles are values of s that cause $G(s) \rightarrow \infty$

$$s = -1 \quad \text{and} \quad s = -3$$

Zeros are values of s that cause $G(s) \rightarrow 0$

$$s = -5$$

Gain is the value of $G(s)$ under steady state conditions

$$G(0) = \frac{0 + 5}{(0 + 1)(0 + 3)} = \frac{5}{3}$$

Poles and Zeros

- ❑ Pole due to input -> forced response
- ❑ Pole due to transfer function -> natural response
- ❑ Zero and pole both contribute to the amplitudes of the response

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau$$

$$= \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t-\tau)\mathbf{B}\mathbf{u}(\tau)d\tau$$

zero-input response

Zero-state response as
convolution integral

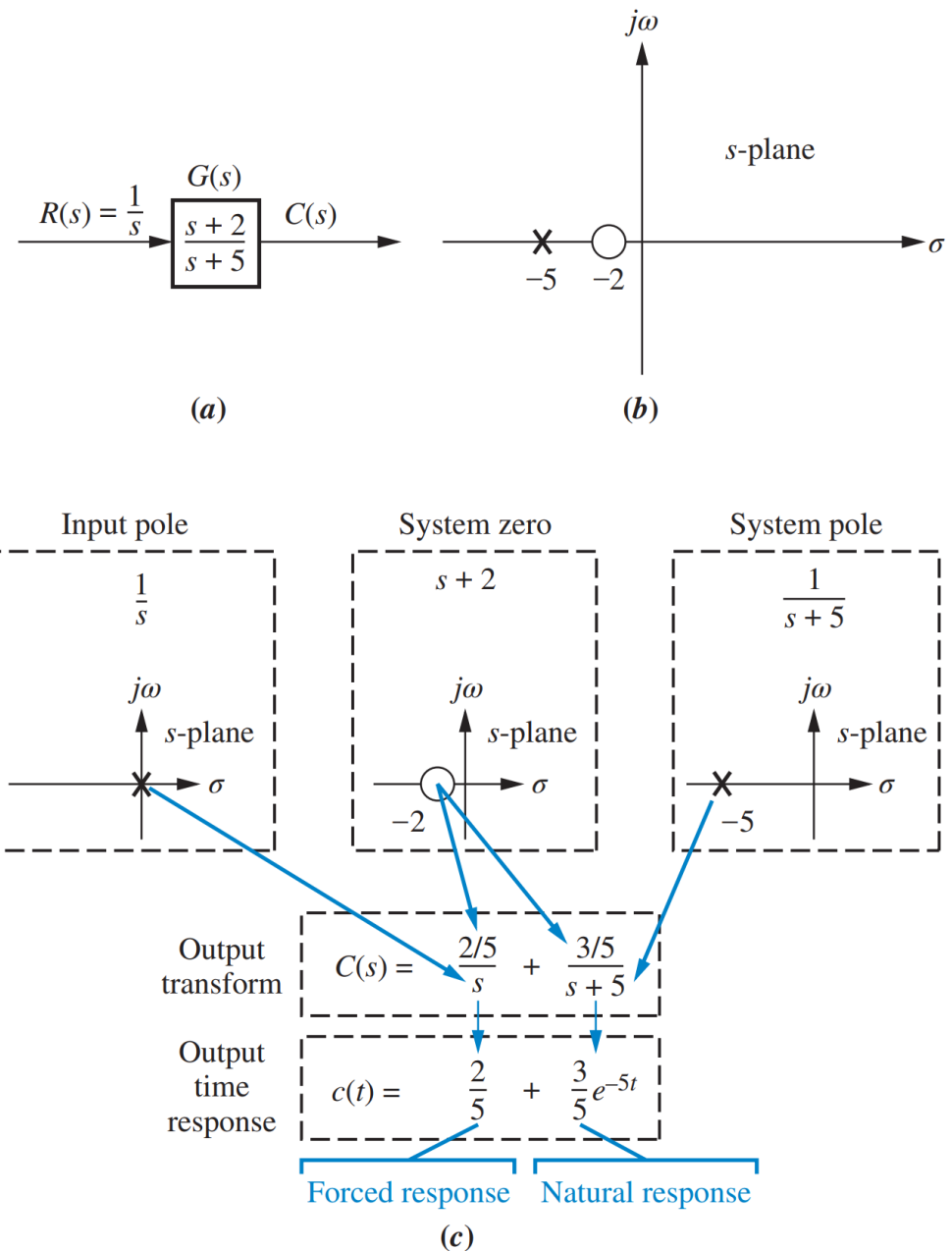


FIGURE 4.1 a. System showing input and output; b. pole-zero plot of the system; c. evolution of a system response. Follow blue arrows to see the evolution of the response component generated by the pole or zero.

Evaluating Response Using Poles

PROBLEM: Given the system of Figure 4.3, write the output, $c(t)$, in general terms. Specify the forced and natural parts of the solution.

SOLUTION: By inspection, each system pole generates an exponential as part of the natural response. The input's pole generates the forced response. Thus,

$$C(s) \equiv \underbrace{\frac{K_1}{s}}_{\text{Forced response}} + \underbrace{\frac{K_2}{s+2} + \frac{K_3}{s+4} + \frac{K_4}{s+5}}_{\text{Natural response}} \quad (4.3)$$

Taking the inverse Laplace transform, we get

$$c(t) \equiv \underbrace{K_1}_{\text{Forced response}} + \underbrace{K_2 e^{-2t} + K_3 e^{-4t} + K_4 e^{-5t}}_{\text{Natural response}} \quad (4.4)$$

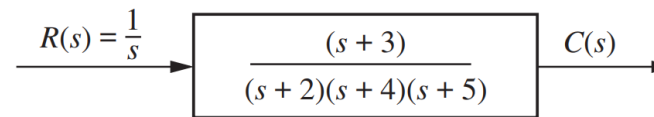


FIGURE 4.3 System for Example 4.1

PROBLEM: A system has a transfer function, $G(s) = \frac{10(s+4)(s+6)}{(s+1)(s+7)(s+8)(s+10)}$.

Write, by inspection, the output, $c(t)$, in general terms if the input is a unit step.

ANSWER: $c(t) \equiv A + Be^{-t} + Ce^{-7t} + De^{-8t} + Ee^{-10t}$



Typical Systems (excited with test signal)

- Step response of the first order system (with no zero)
 - Forced and natural response:

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$

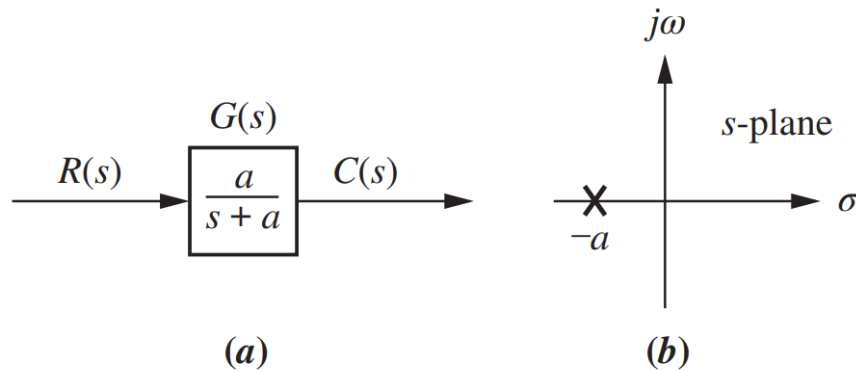


FIGURE 4.4 a. First-order system; b. pole plot

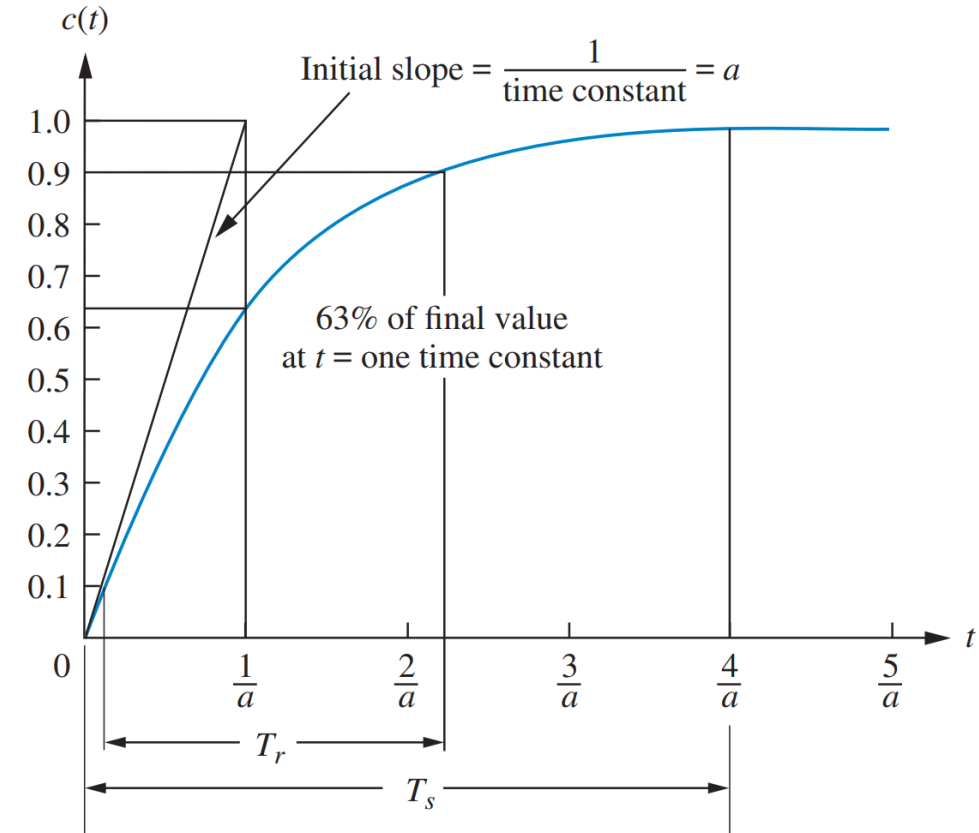


FIGURE 4.5 First-order system response to a unit step

second order system

Step response of the second order system (with no zero) depends on pole locations on the pzmap:

- How about right plane pole?

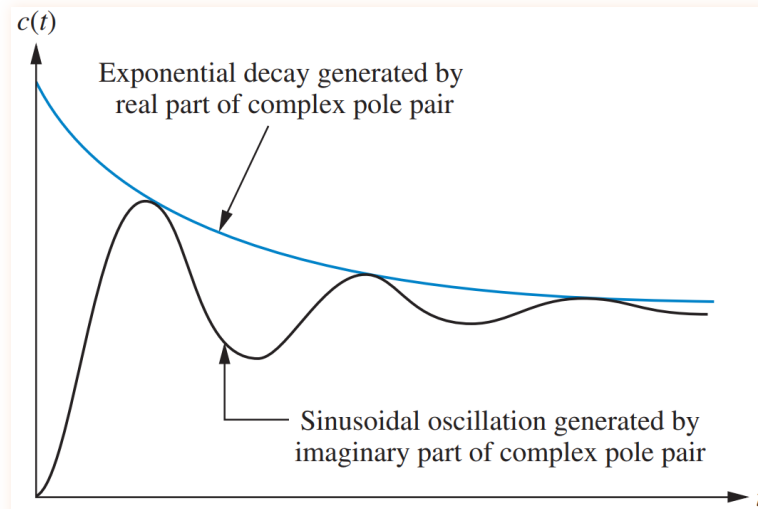


FIGURE 4.8 Second-order step response components generated by complex poles

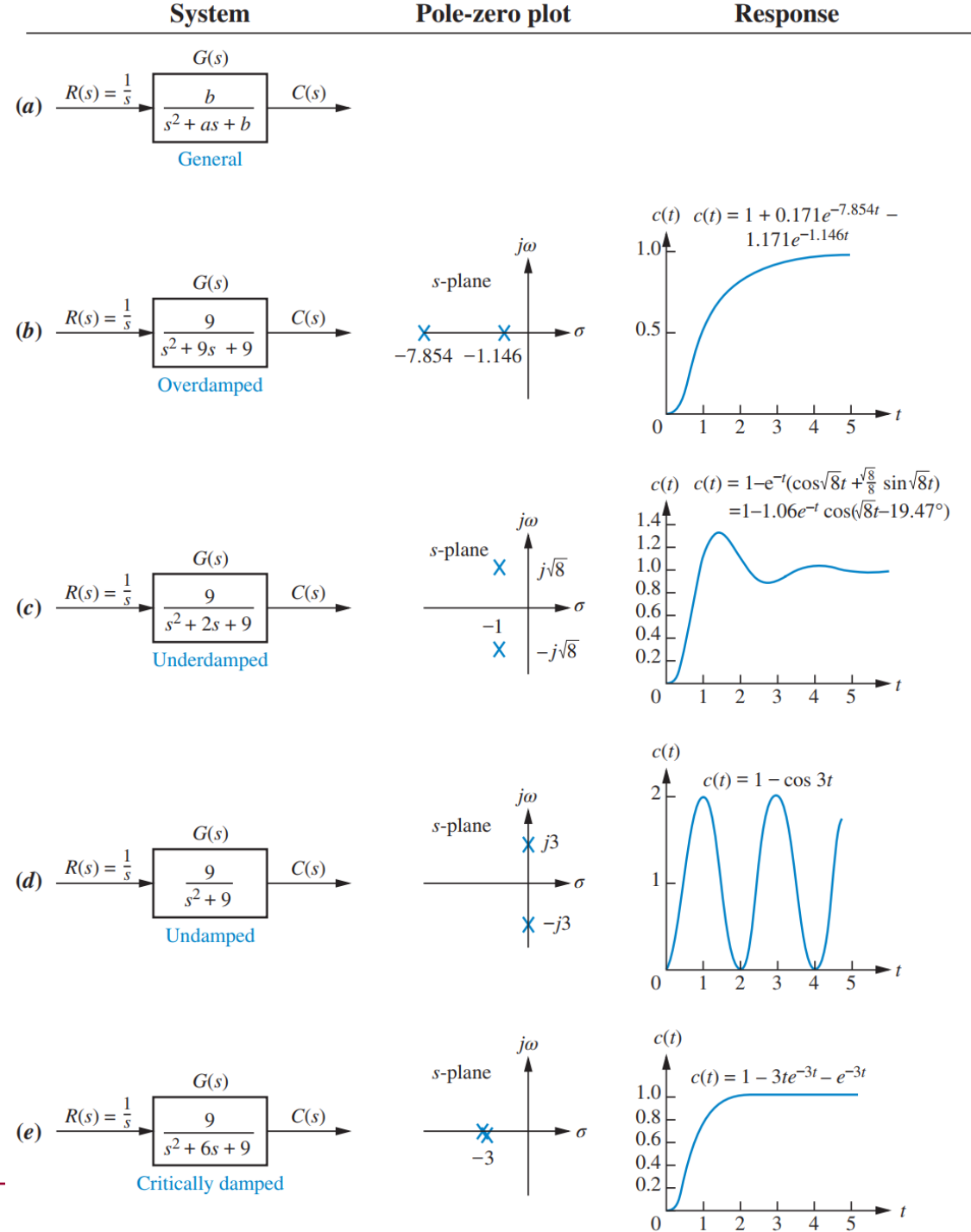
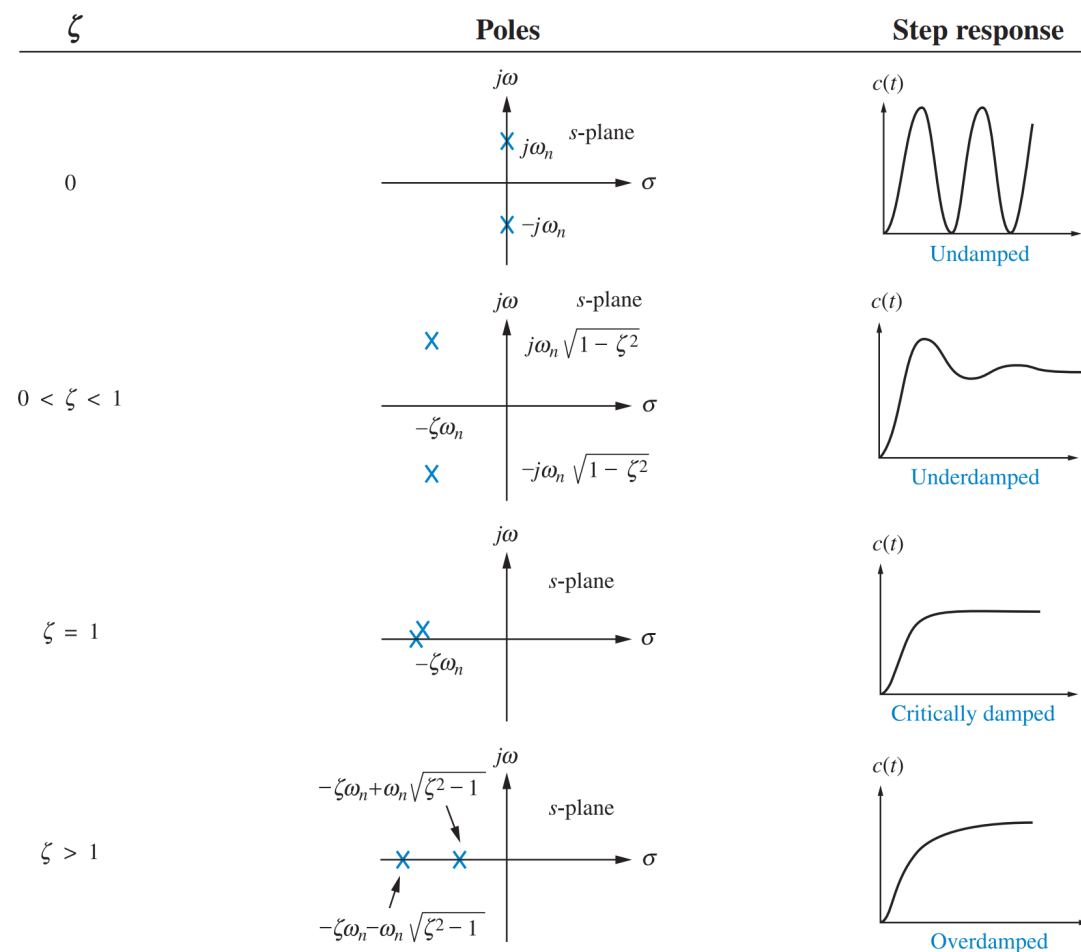


FIGURE 4.7 Second-order systems, pole plots, and step responses

- ❑ Natural frequency ω_n defines the time scale of the time response plot
- ❑ Damping ratio ζ defines the shape of the time response plot

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



Finding ζ and ω_n For a Second-Order System

PROBLEM: Given the transfer function of Eq. (4.23), find ζ and ω_n .

$$G(s) = \frac{36}{s^2 + 4.2s + 36} \quad (4.23)$$

SOLUTION: Comparing Eq. (4.23) to (4.22), $\omega_n^2 = 36$, from which $\omega_n = 6$. Also, $2\zeta\omega_n = 4.2$. Substituting the value of ω_n , $\zeta = 0.35$.

FIGURE 4.11 Second-order response as a function of damping ratio

- Time domain performance metrics are defined in terms of the underdamped response of a second order system with $\zeta \in (0,1)$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$\begin{aligned} c(t) &= 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1-\zeta^2} t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} t \right) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t - \phi) \end{aligned}$$

where $\phi = \tan^{-1}(\zeta/\sqrt{1-\zeta^2})$.

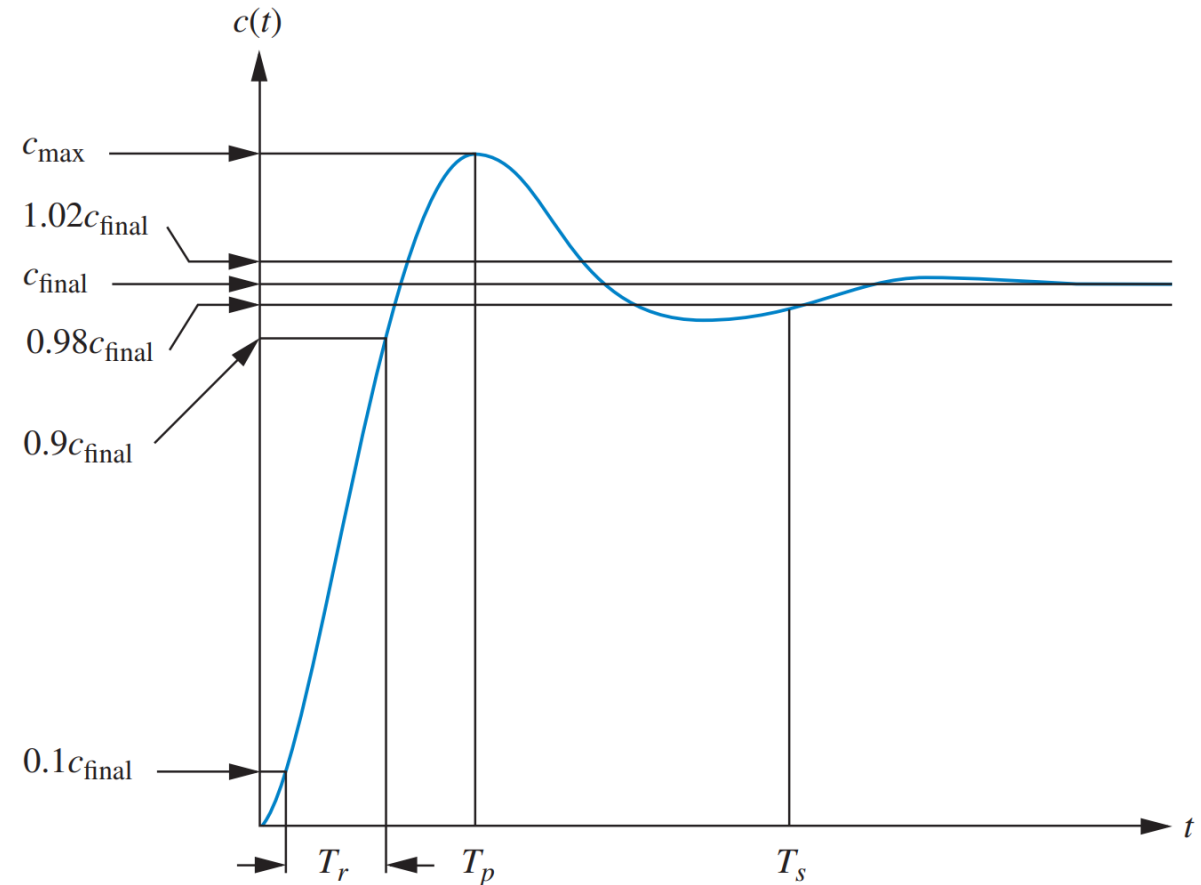


FIGURE 4.14 Second-order underdamped response specifications

Finding T_p , %OS, T_s , and T_r from a Transfer Function

PROBLEM: Given the transfer function

$$G(s) = \frac{100}{s^2 + 15s + 100} \quad (4.43)$$

find T_p , %OS, T_s , and T_r .

SOLUTION: ω_n and ζ are calculated as 10 and 0.75, respectively. Now substitute ζ and ω_n into Eqs. (4.34), (4.38), and (4.42) and find, respectively, that $T_p = 0.475$ second, %OS = 2.838, and $T_s = 0.533$ second. Using the table in Figure 4.16, the normalized rise time is approximately 2.3 seconds. Dividing by ω_n yields $T_r = 0.23$ second. This problem demonstrates that we can find T_p , %OS, T_s , and T_r without the tedious task of taking an inverse Laplace transform, plotting the output response, and taking measurements from the plot.

Time domain performance metrics

□ Peak time and settling time are related to :

- damped frequency of oscillation, ω_d
- and exponential damping frequency σ_d

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\zeta \omega_n} = \frac{\pi}{\sigma_d}$$

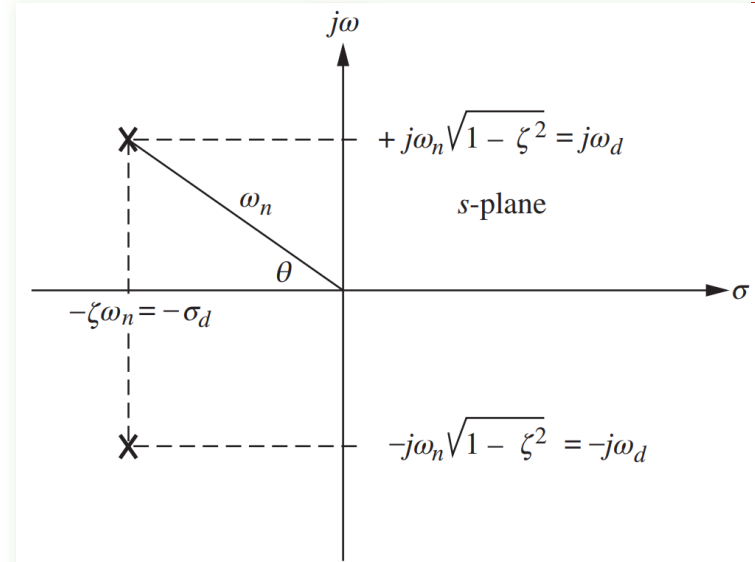


FIGURE 4.17 Pole plot for an underdamped second-order system

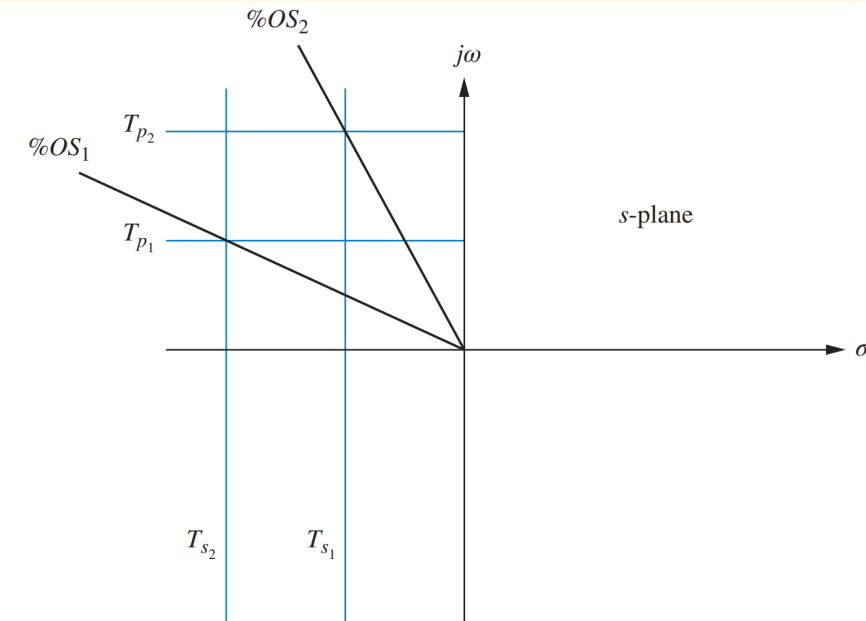


FIGURE 4.18 Lines of constant peak time, T_p , settling time, T_s , and percent overshoot, %OS. Note: $T_{s2} < T_{s1}$; $T_{p2} < T_{p1}$; $\%OS_1 < \%OS_2$.

Example responses

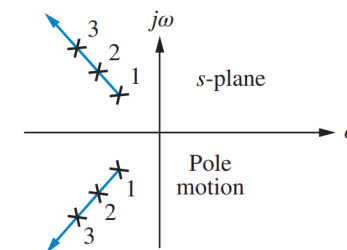
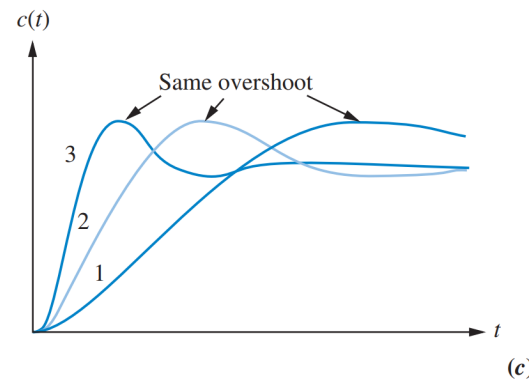
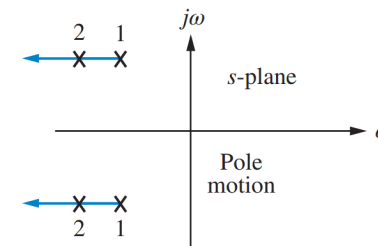
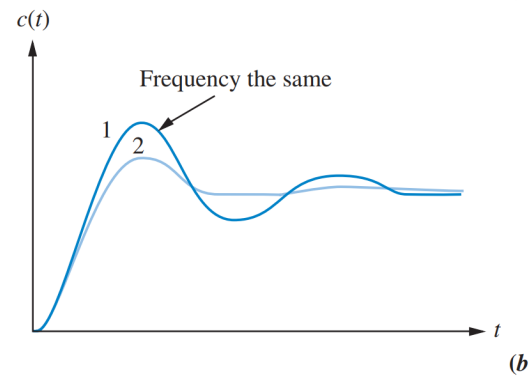
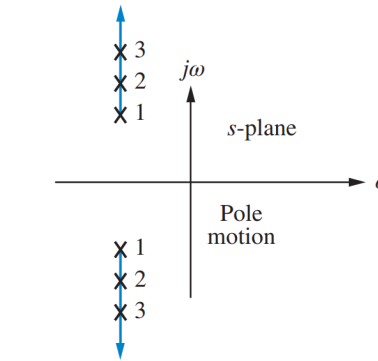
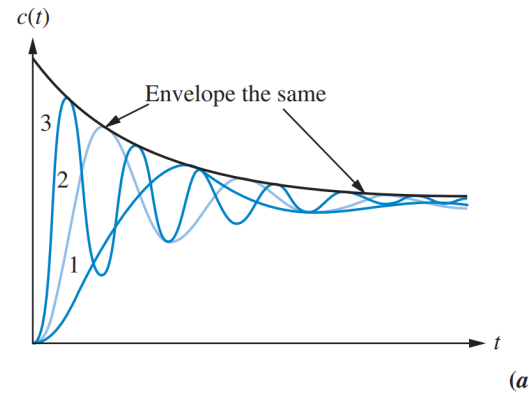


FIGURE 4.19 Step responses of second-order underdamped systems as poles move: **a.** with constant real part; **b.** with constant imaginary part; **c.** with constant damping ratio

Finding T_p , %OS, and T_s from Pole Location

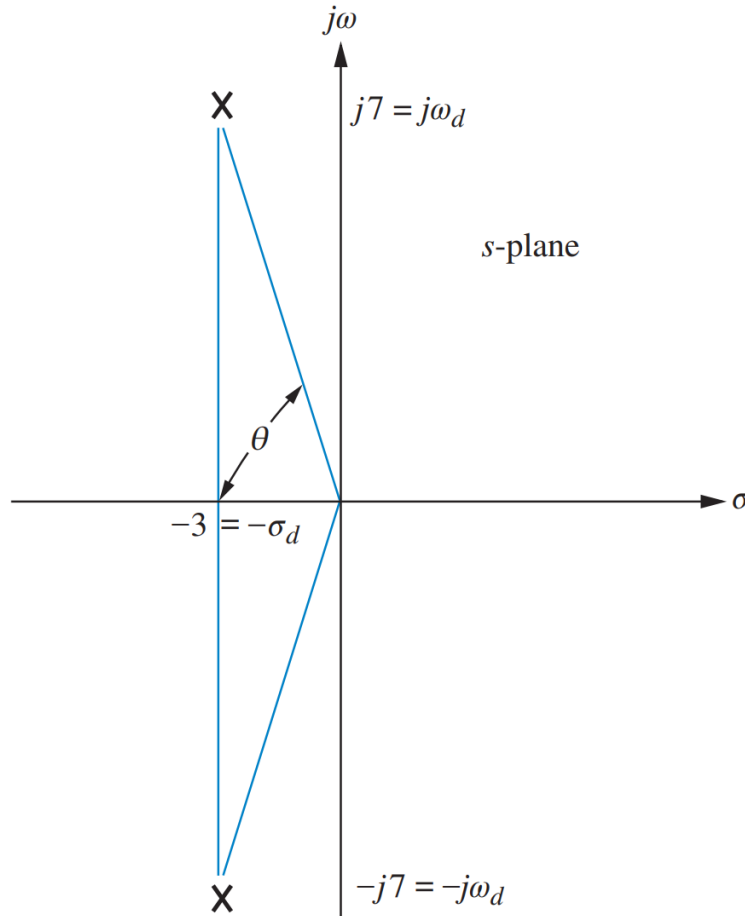


FIGURE 4.20 Pole plot for Example 4.6

PROBLEM: Given the pole plot shown in Figure 4.20, find ζ , ω_n , T_p , %OS, and T_s .

SOLUTION: The damping ratio is given by $\zeta = \cos \theta = \cos[\arctan(7/3)] = 0.394$. The natural frequency, ω_n , is the radial distance from the origin to the pole, or $\omega_n = \sqrt{7^2 + 3^2} = 7.616$. The peak time is

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{7} = 0.449 \text{ second} \quad (4.46)$$

The percent overshoot is

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 26\% \quad (4.47)$$

The approximate settling time is

$$T_s = \frac{4}{\sigma_d} = \frac{4}{3} = 1.333 \text{ seconds} \quad (4.48)$$

Dominant Pole Concept



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Response of systems with zeros



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Stability

(in terms of Routh-Hurwitz Criterion)